The model, and what it is checked against

This is the evidence behind the model cards in libs.tech/ngspice/igzo_mmm_lab.ngspice: what was measured, what was fitted to it, how close the fit comes, and - the part that matters for planning the next run - which measurement would remove which uncertainty.

Every figure here is generated from the raw .xls files in measurements/ and from ngspice runs against the PDK's own corner sections. Nothing is copied from an earlier report, so nothing can quietly go stale.

What the model is

A SPICE level-1 MOSFET with three things wrapped around it, in the igzo_tft subcircuit:

contact resistance   Rc = 3.3 Ohm*m / W per contact, measured
overlap capacitance Cox * ov * W per side, from the drawn overlap
a bulk that stays inside - a TFT has no body terminal

Level 1 was chosen because the data supports four numbers per corner and not more: Vto, Kp, lambda and Tox. A model with more parameters than the measurement can pin is a model that fits noise.

The four corners, and where they come from

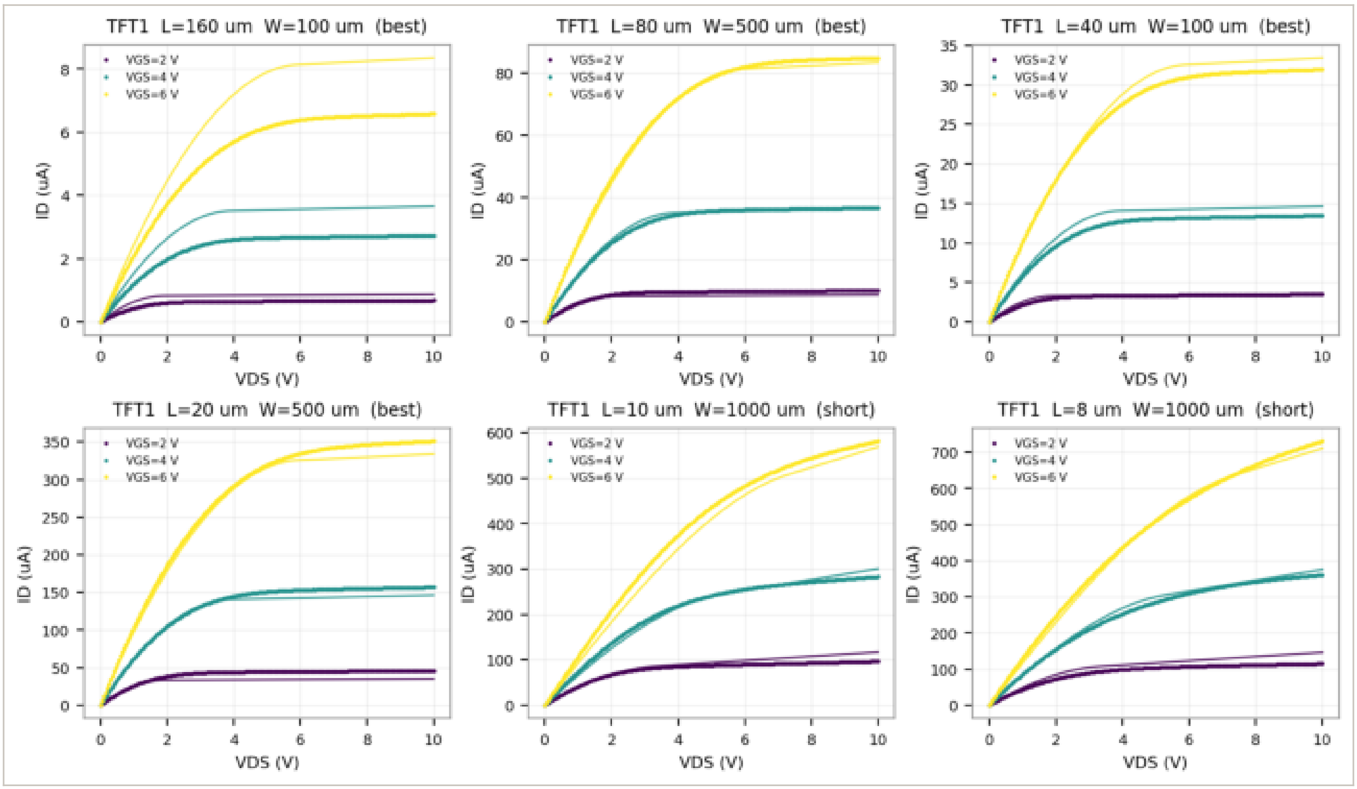
Corner	Vto (V)	Kp (uA/V^2)	lambda (1/V)	What it is
best	+0.09	0.717	0.0067	long-channel devices of the best chip
tt	-0.26	0.562	0.0142	median over every working device, best chip
short	-1.32	0.128	0.0659	L <= 10 um, contact-limited
all	-0.53	0.381	0.0355	all three chips, the bad ones included

Where this model should not be trusted

1. It is DC-only. It reproduces the curves above and nothing else has been checked against silicon.
2. Valid over the measured range: VGS <= 6 V, VDS <= 10 V. Outside it the square law is extrapolation, not prediction.
3. No temperature, no bias stress, no hysteresis, no noise model.
4. The gate capacitance is derived, not measured, so anything that depends on charging the gate - fT, tuned loads, switching speed - is an estimate resting on a 5 um overlap read off a drawing.

Output curves: simulated over measured

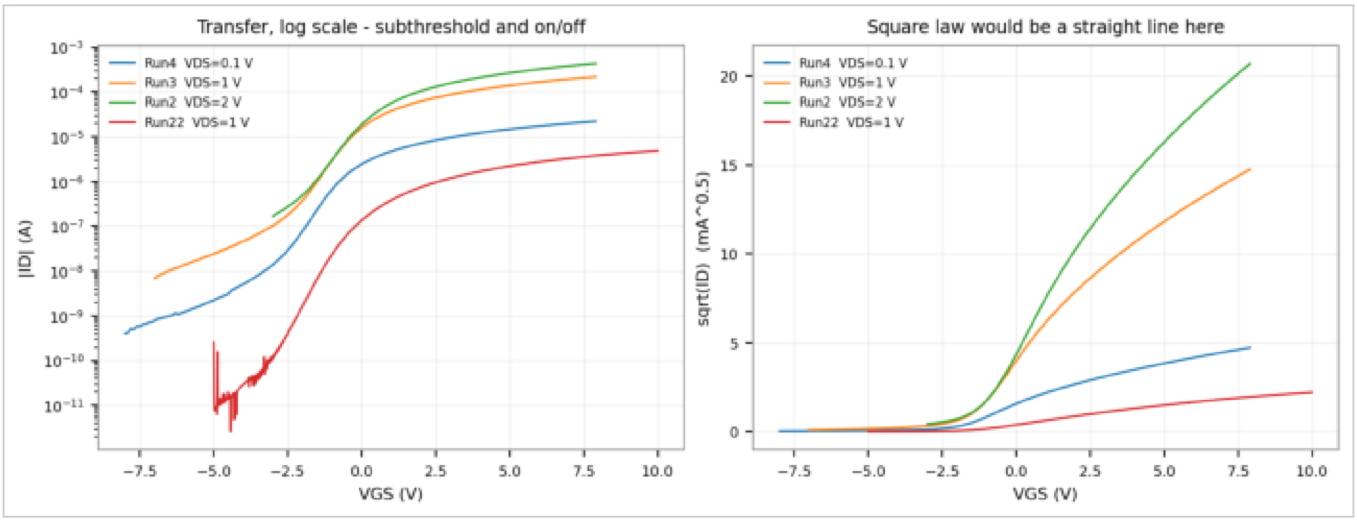
Points are measured, lines are ngspice at the same W and L. Six devices spanning L = 8 to 160 um, at every gate voltage the sweep carries.



Median simulated/measured in saturation: 1.01 over 18 comparisons.

Transfer curves: where the square law gives out

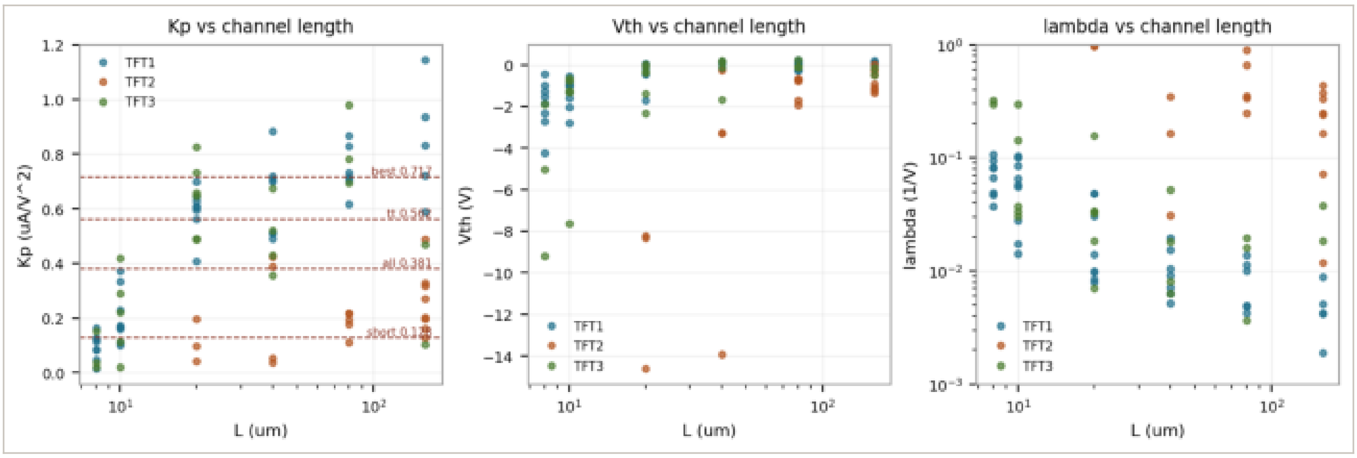
The left panel is the log sweep - subthreshold slope and on/off ratio, neither of which a level-1 model reproduces at all. The right panel is sqrt(ID): a square-law device is a straight line there, and these are not. The measured saturation exponent is 2.31 rather than 2, which is why the model runs about 20 % pessimistic on gm at a given current.



run	VDS (V)	Kp*W/L (uA/V^2)	Vth (V)	S (V/dec)	on/off
Run4	0.1	26.4	-0.10	2.22	5e+04
Run3	1	27.1	-0.00	2.16	3e+04
Run2	2	28.4	0.52	1.36	3e+03
Run22	1	0.554	5.56	1.53	2e+06

Every device fit, and the corners drawn from them

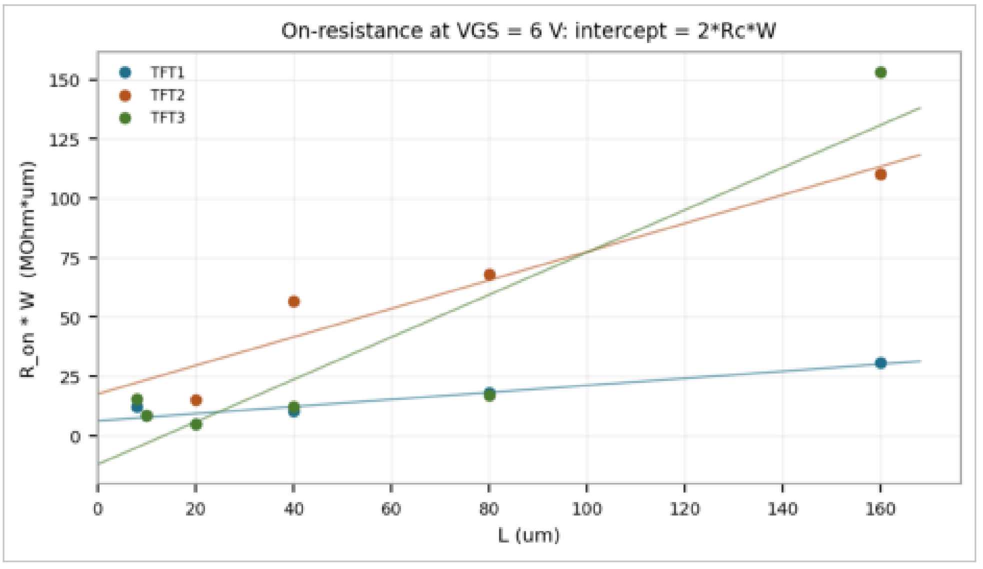
86 devices, three chips. The spread is the point: Kp varies by a factor of 70 across the population and falls off sharply below L = 20 um, which is the contact resistance taking over rather than the mobility changing. TFT2 sits apart from the other two - its saturation exponent comes out at 0.87, which no square-law model can express - and it is excluded from the tt and best corners for that reason, and only for that reason.



Contact resistance, from the devices themselves

R_on * W against L at VGS = 6 V. The slope is the gated sheet resistance and the intercept is 2*Rc*W = 6.6 MOhm*um. This is the primary route rather than TLM because the TLM pad spacing was never recorded in the .xls files; the TLM ladders are in the data and are reported as measured, but they cannot be turned into an absolute Rc without that number.

The three chips do not agree, and the table says so: 6.6 MOhm*um is the best chip. The last column is the length below which the contacts dominate the device - which is why the short corner exists.

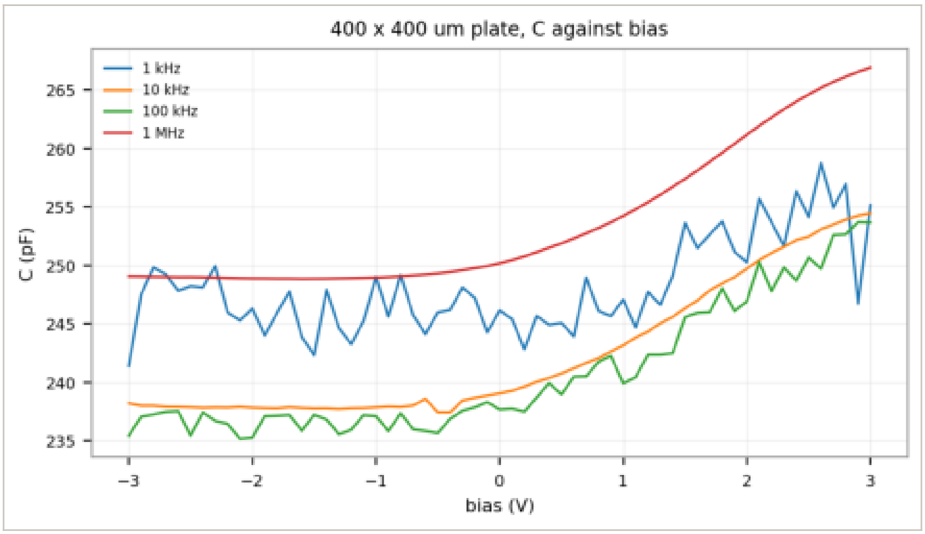


Chip	2*Rc*W (MOhm*um)	sheet R (MOhm*um / um)	contact-limited below
TFT1	6.36	0.149	43
TFT2	17.61	0.599	29
TFT3	-12.01	0.892	-13

Cox, and exactly what it does and does not establish

The 400 x 400 um plate of Test_capacitance, swept from -3 to +3 V at four frequencies. At 1 MHz it reads 250.2 pF, which over 160 000 um^2 gives Cox = 1.564 fF/um^2 - the one capacitance in this PDK that is measured rather than derived.

What it does not establish: the transistor's gate capacitance. This is a metal-insulator-metal plate, not a gate over a channel, so it fixes the dielectric and nothing about how the channel charges and discharges. The overlap capacitance in the model is Cox times an overlap read off one GDS cell, and no measurement anywhere in this data set confirms it.



What to measure next, and what each measurement buys

Measurement	What it would settle
C-V on a transistor, gate to S/D	the largest gap: Cgs and Cgd against bias, which the model has never seen
A proper VGS sweep per chip	Vth and subthreshold slope on all three chips - the VGS folders are empty
TLM with the pad spacing recorded	an absolute Rc, independent of the R_on fit that currently carries it
Drawn vs printed across a wafer	the +3 um bias is one cell; its spread is what a PCell would need
Bias stress and hysteresis	IGZO drifts under gate bias; the model has no term for it at all
Metal thickness, gate and S/D	every inductor resistance in this PDK is an assumption until this exists
S-parameters or a ring oscillator	fT, and whether the DC model means anything at 100 MHz